

Brac University
Department of Electrical & Electronic Engineering
Spring2026

Course Number : EEE308L

Section :06

Group No :04



Lab Report

Experiment no: 06

Name of the experiment:

Study of Operational Amplifier as Wien Bridge Oscillator.

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Experiment 06: Study of Operational Amplifier as Wien Bridge Oscillator.

Objective: The objective of this experiment is to study the operation of the Wien bridge oscillator.

Theoretical Background:

The Wien Bridge Oscillator is a circuit that generates a stable sine wave output by using an operational amplifier paired with a bridge network of resistors and capacitors. This bridge acts as a feedback loop that determines the specific frequency of the wave

$$f = \frac{1}{2\pi RC}$$

For the circuit to work effectively, the amplifier's gain must be carefully balanced: too much gain causes the signal to distort or "clip," while too little gain prevents the oscillation from sustaining. In this experiment, the setup successfully produced clear signals for testing by maintaining this feedback balance.

Equipment:

- Op-amp (741)-1 piece
- Resistor: 10 k Ω -4 pieces
- Resistor: 33 k Ω -1 piece
- Capacitor 0.22 μ F -2 pieces
- DC power supply -1 unit(+15V, -15V)
- Oscilloscope -1 unit
- Analog Trainer Board -1 unit
- Chords and wire

Circuit Diagram:

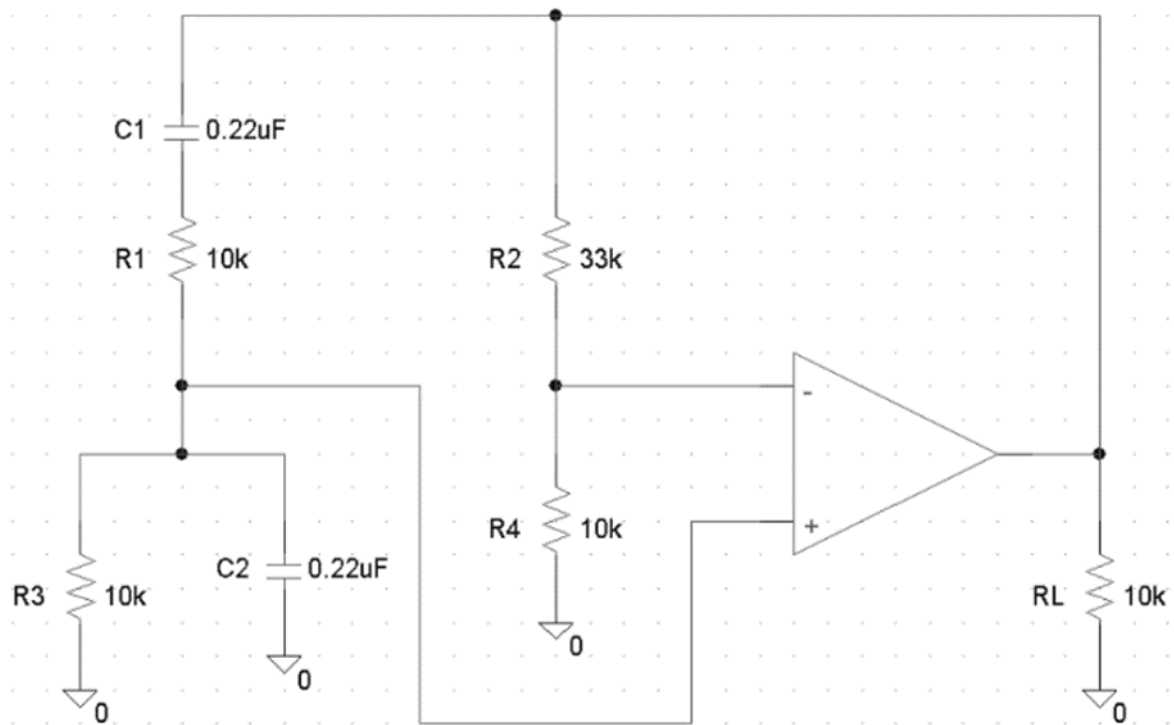


Fig: Wien Bridge Oscillator circuit

SETUP AND OUTPUT:

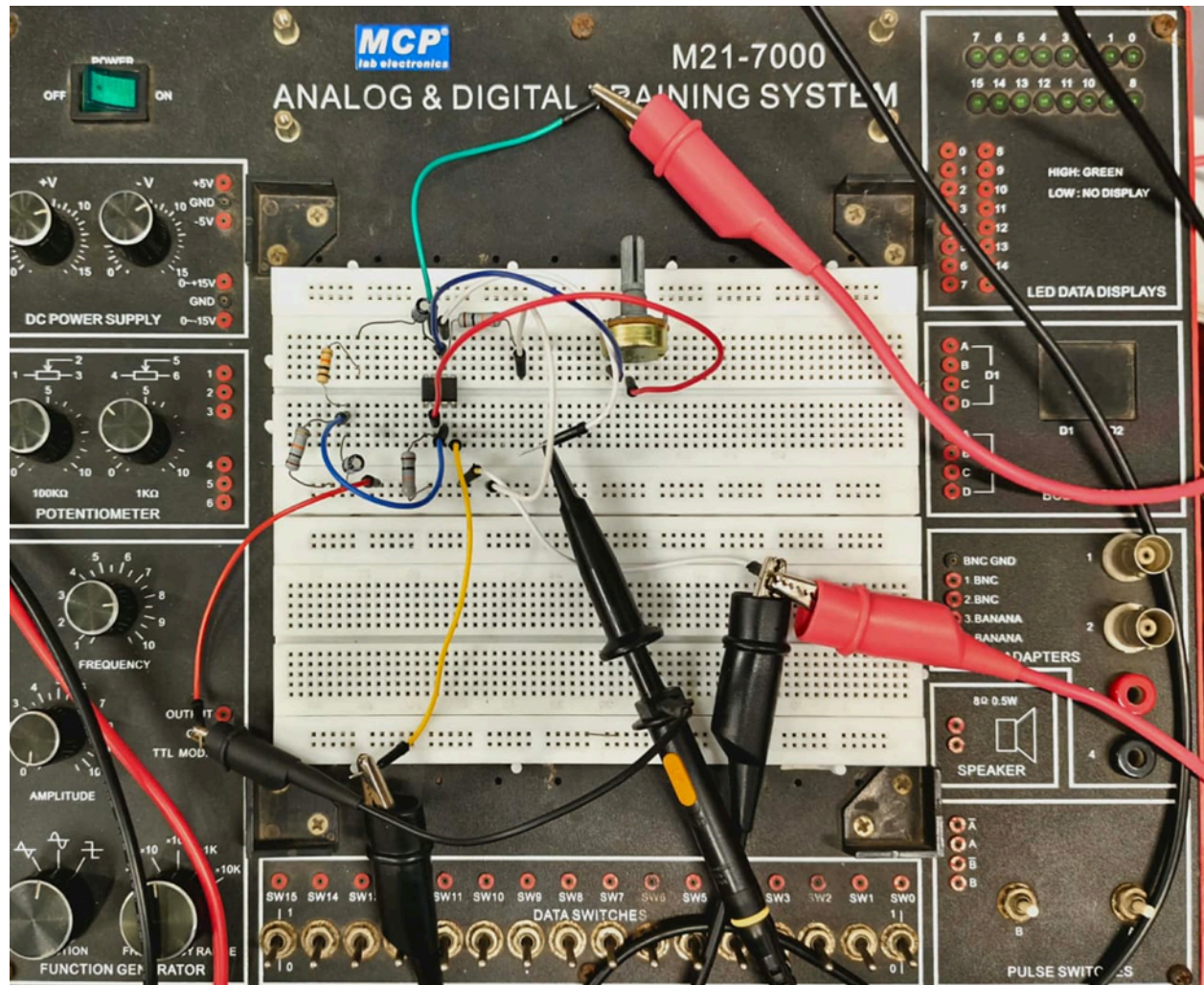


Fig: Hardware setup for Wien Bridge Oscillator circuit

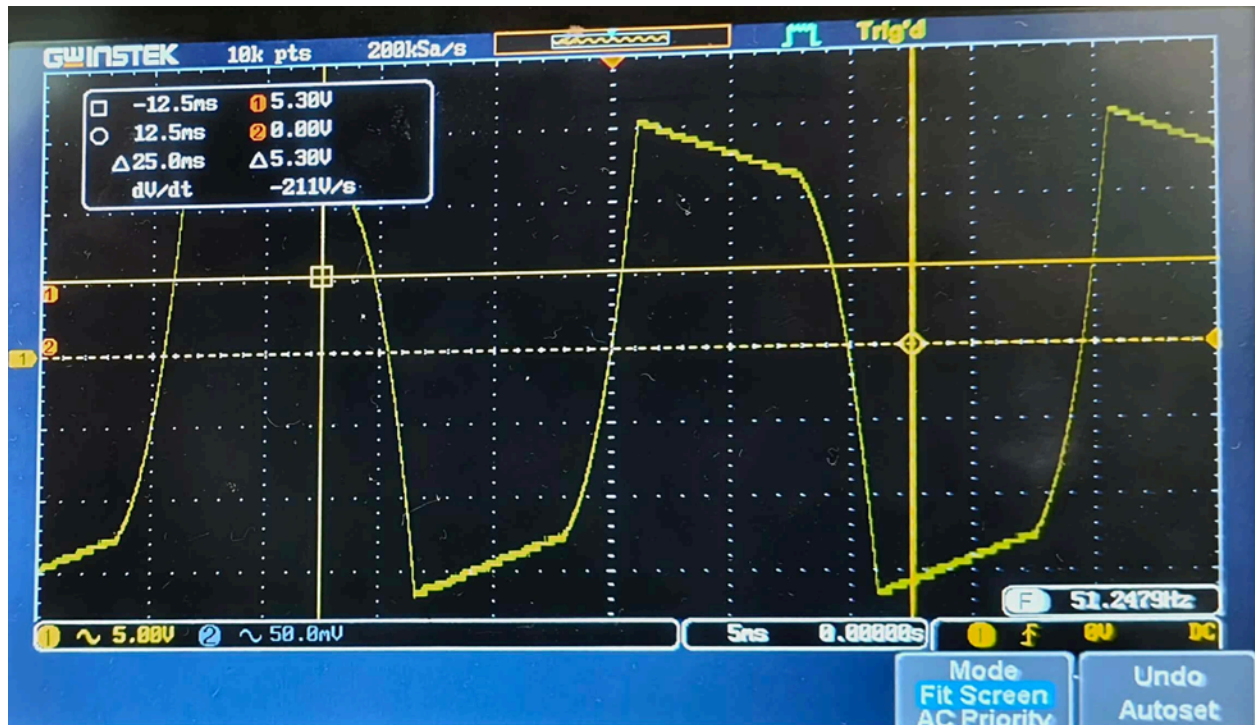


Fig: Output for 33K

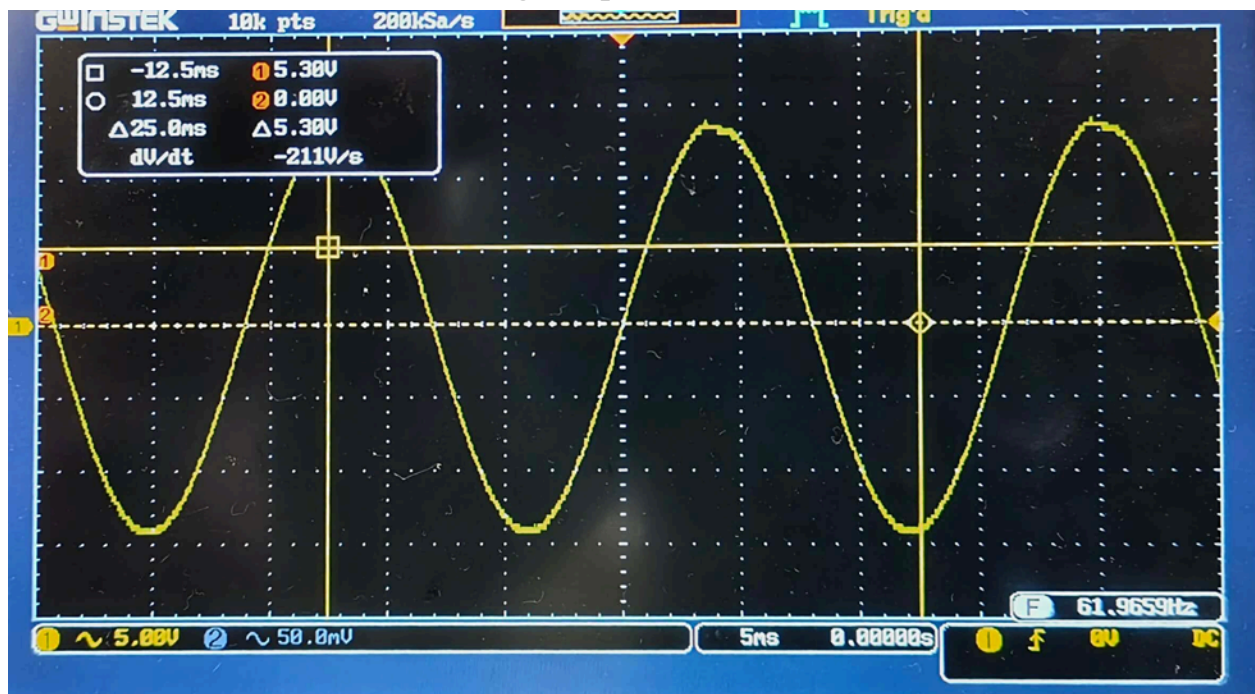


Fig: Output for 20K

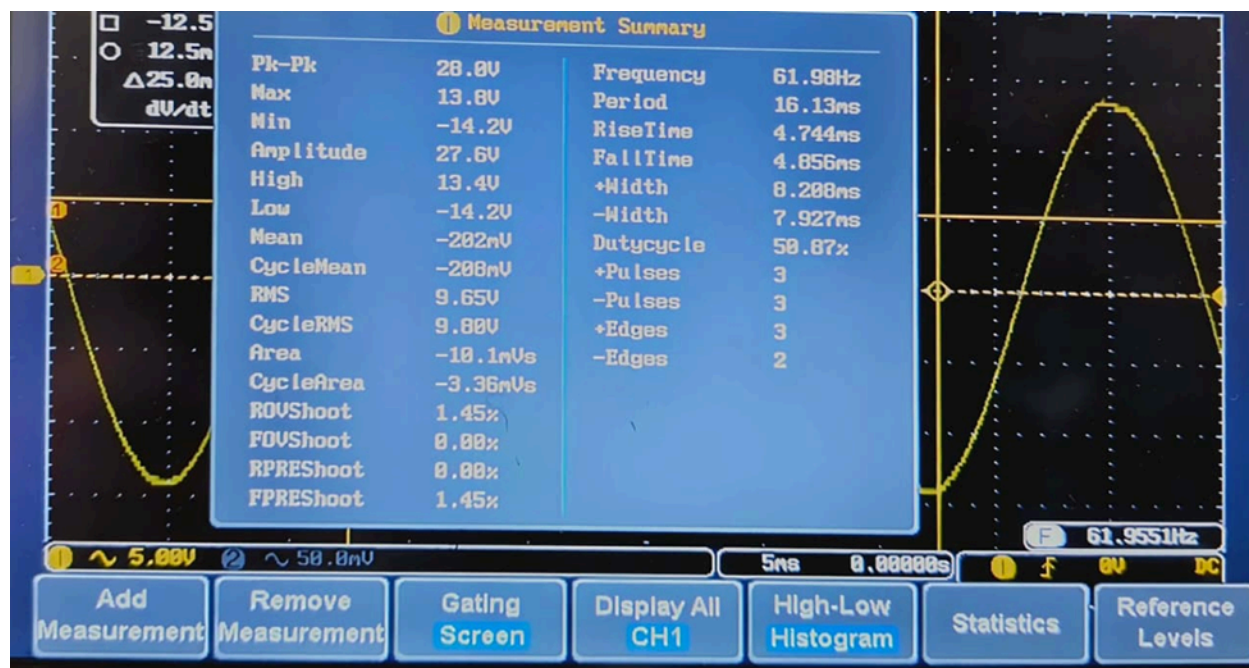


Fig: Measurement of the output waveform

Pspice Simulation:-

Circuit Diagram:

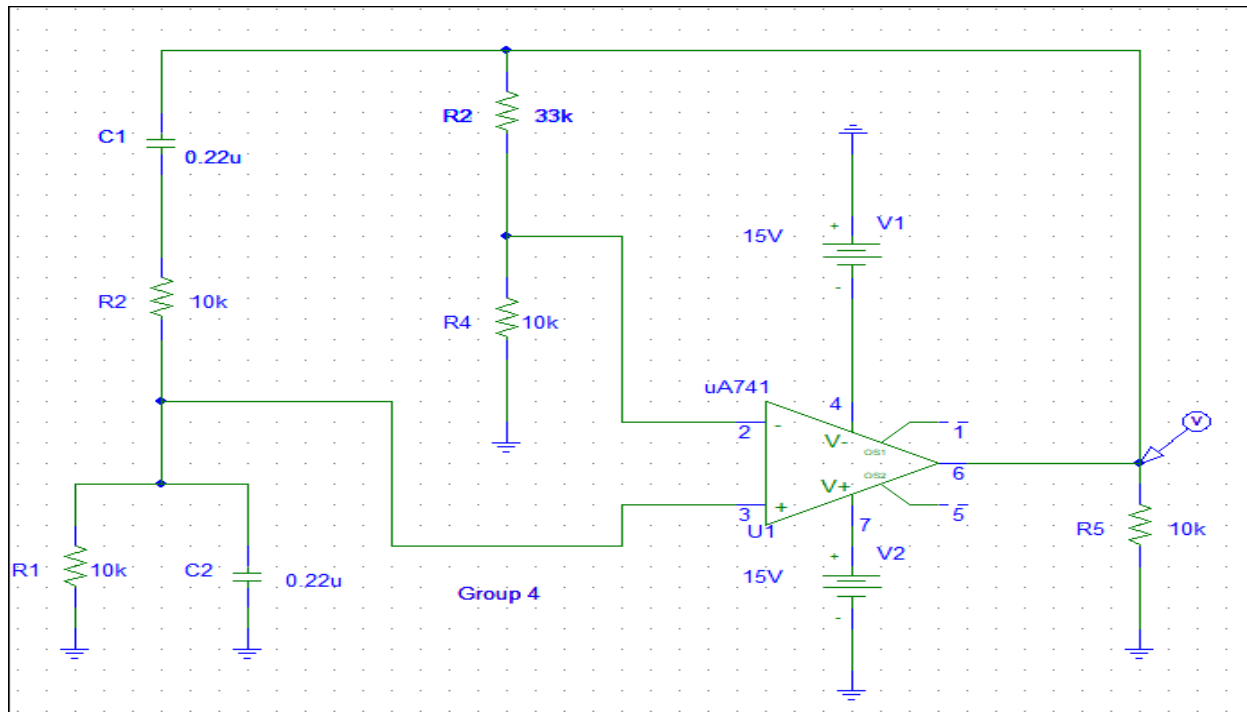


Fig: Software setup for Wien Bridge Oscillator circuit [$R_2=33k$]

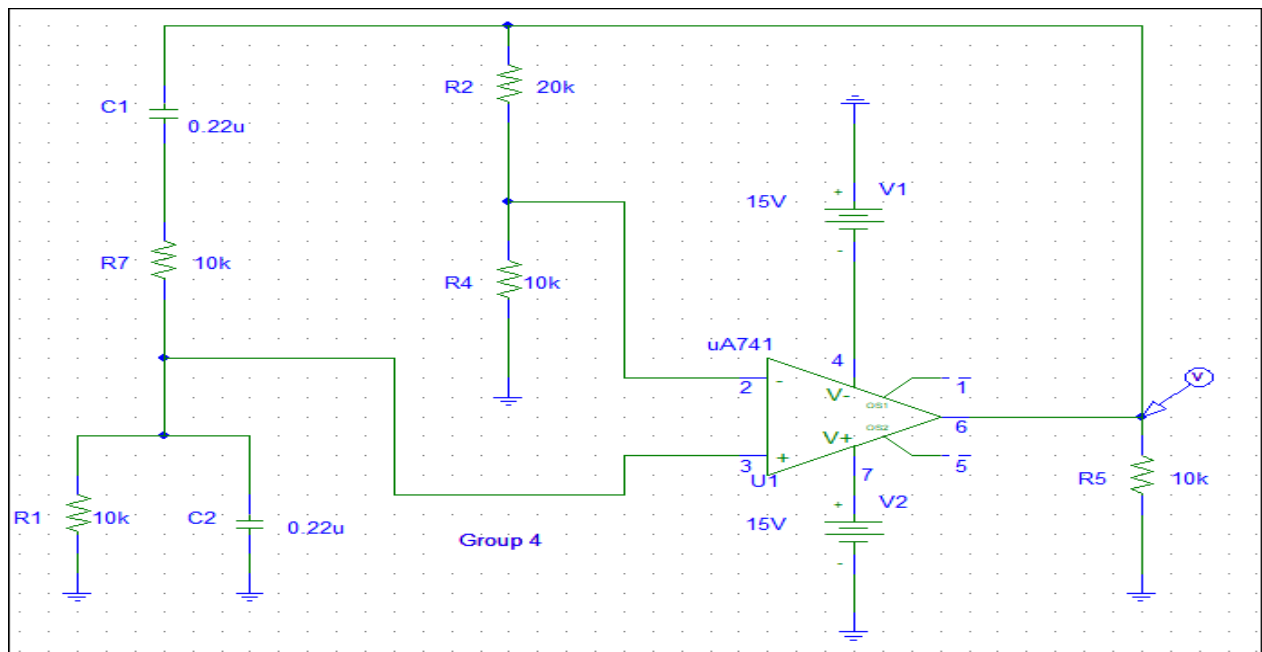
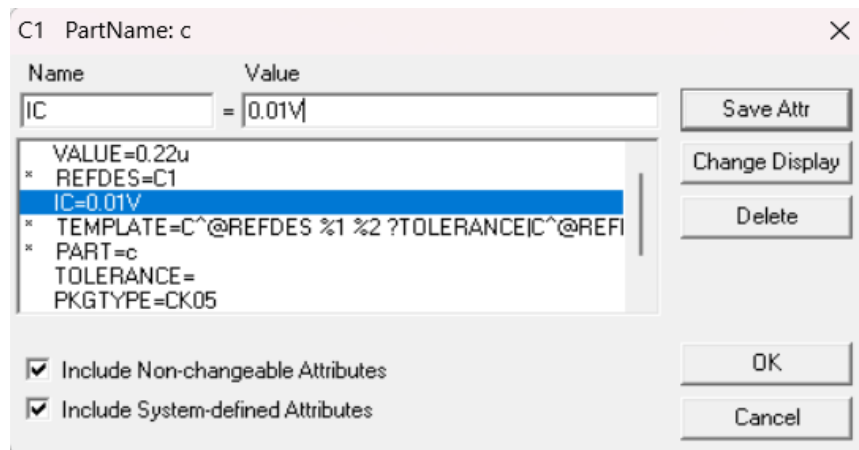


Fig: Software setup for Wien Bridge Oscillator circuit [$R_2=20k$]

C1 Parameter:



A dialog box titled "C1 PartName: c" with a close button (X) in the top right corner. It contains a table with two columns: "Name" and "Value". The first row shows "IC" in the Name column and "= 0.01V" in the Value column. Below the table is a list of attributes: "VALUE=0.22u", "* REFDES=C1", "IC=0.01V" (which is highlighted in blue), "* TEMPLATE=C^@REFDES %1 %2 ?TOLERANCE|C^@REFI", "* PART=c", "TOLERANCE=", and "PKGTYPE=CK05". To the right of the list are three buttons: "Save Attr", "Change Display", and "Delete". At the bottom left are two checked checkboxes: "Include Non-changeable Attributes" and "Include System-defined Attributes". At the bottom right are two buttons: "OK" and "Cancel".

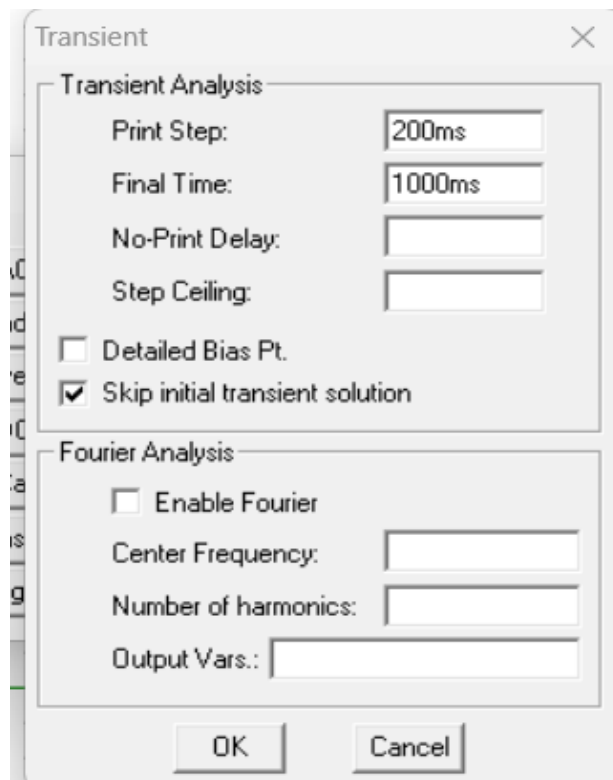
Name	Value
IC	= 0.01V

VALUE=0.22u
* REFDES=C1
IC=0.01V
* TEMPLATE=C^@REFDES %1 %2 ?TOLERANCE|C^@REFI
* PART=c
TOLERANCE=
PKGTYPE=CK05

☒ Include Non-changeable Attributes
☒ Include System-defined Attributes

Save Attr
Change Display
Delete
OK
Cancel

Transient parameter:



A dialog box titled "Transient" with a close button (X) in the top right corner. It has two main sections: "Transient Analysis" and "Fourier Analysis". The "Transient Analysis" section contains: "Print Step:" with a text box containing "200ms", "Final Time:" with a text box containing "1000ms", "No-Print Delay:" with an empty text box, "Step Ceiling:" with an empty text box, an unchecked checkbox for "Detailed Bias Pt.", and a checked checkbox for "Skip initial transient solution". The "Fourier Analysis" section contains: an unchecked checkbox for "Enable Fourier", "Center Frequency:" with an empty text box, "Number of harmonics:" with an empty text box, and "Output Vars.:" with an empty text box. At the bottom are "OK" and "Cancel" buttons.

Transient Analysis

Print Step: 200ms
Final Time: 1000ms
No-Print Delay:
Step Ceiling:
☐ Detailed Bias Pt.
☒ Skip initial transient solution

Fourier Analysis

☐ Enable Fourier
Center Frequency:
Number of harmonics:
Output Vars.:

OK Cancel

Output Graph:-

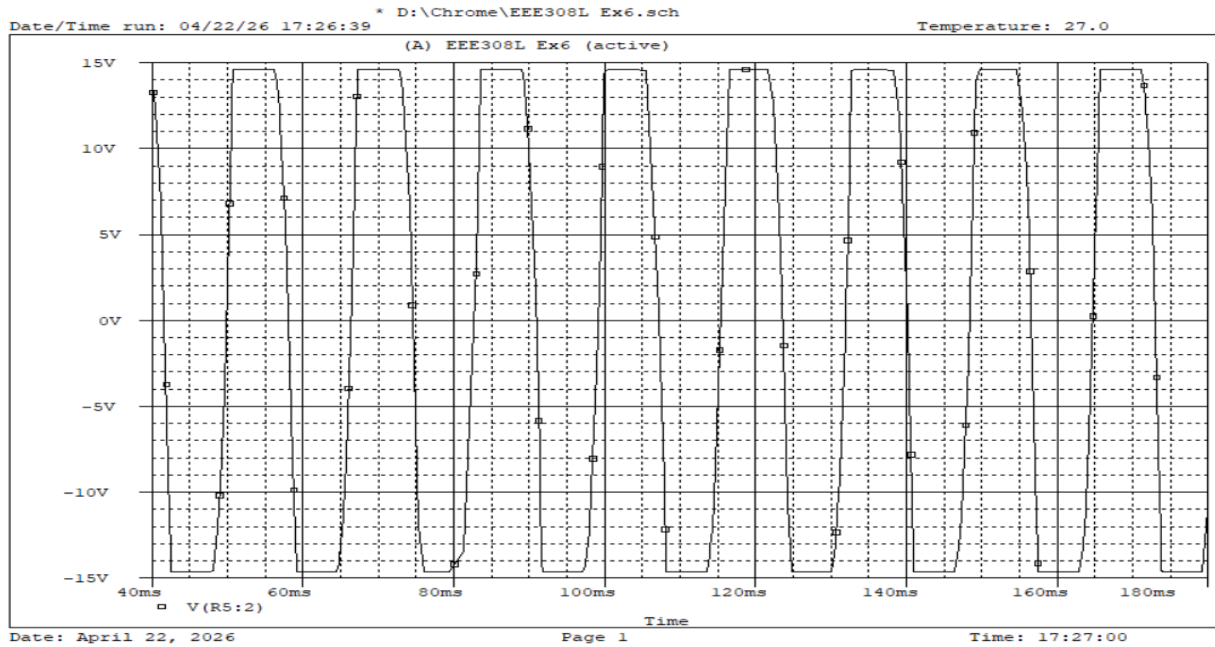


Fig: Graph for 33k

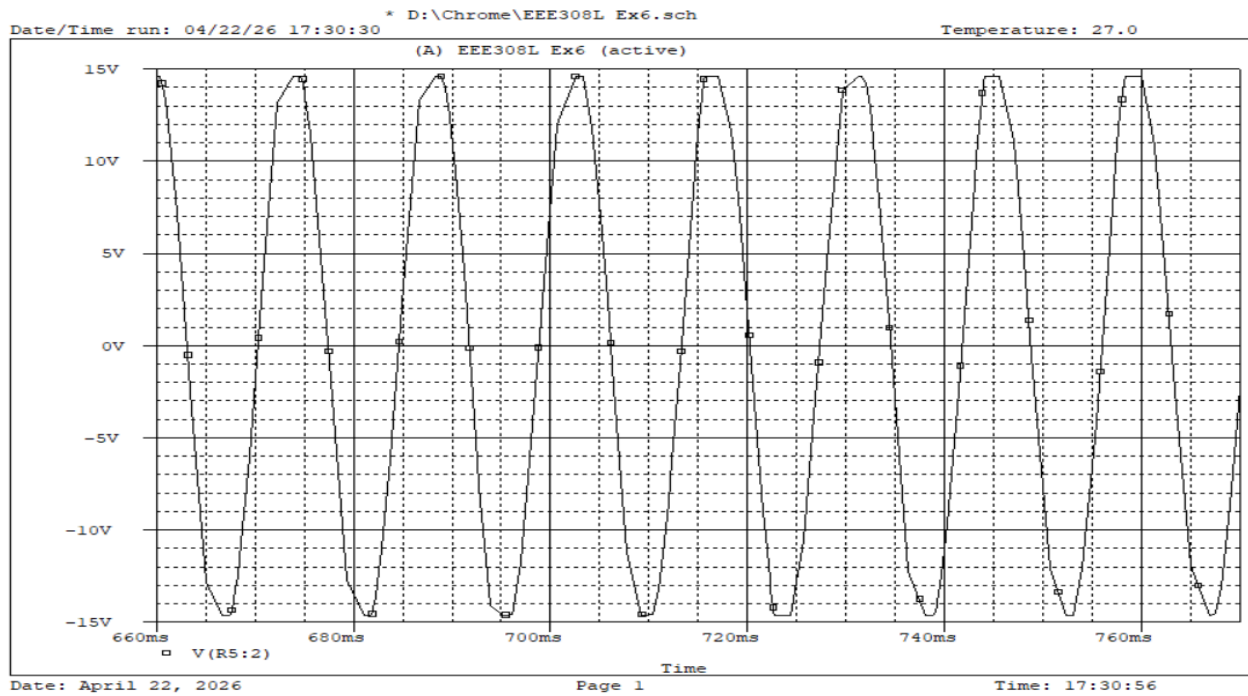


Fig: Graph for 20k

Discussion:

This experiment proved that a Wien Bridge Oscillator can successfully create its own steady signal using a specific feedback loop. We measured a frequency of about 72 Hz, which was almost exactly what we calculated, with only small differences caused by real-world parts not being perfect. The most important lesson was balancing the gain: a perfect setting gave us a smooth, clean wave, while too much gain distorted the signal and too little made it disappear. In the end, the circuit worked just as expected, showing it is a reliable way to generate clear signals for testing and audio equipment

Appendix:-

EEE-308L

Experiment - 6

Group - 4

Wien Bridge Oscillator:

$$R_1 = 10k$$

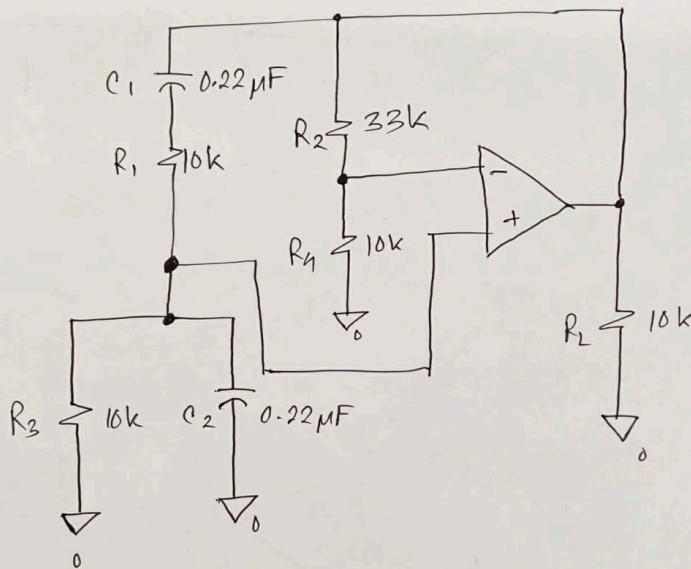
$$R_4 = 9.7k$$

$$R_3 = 9.8k$$

$$R_2 = 32.7k$$

$$C_1 = 0.23\mu F$$

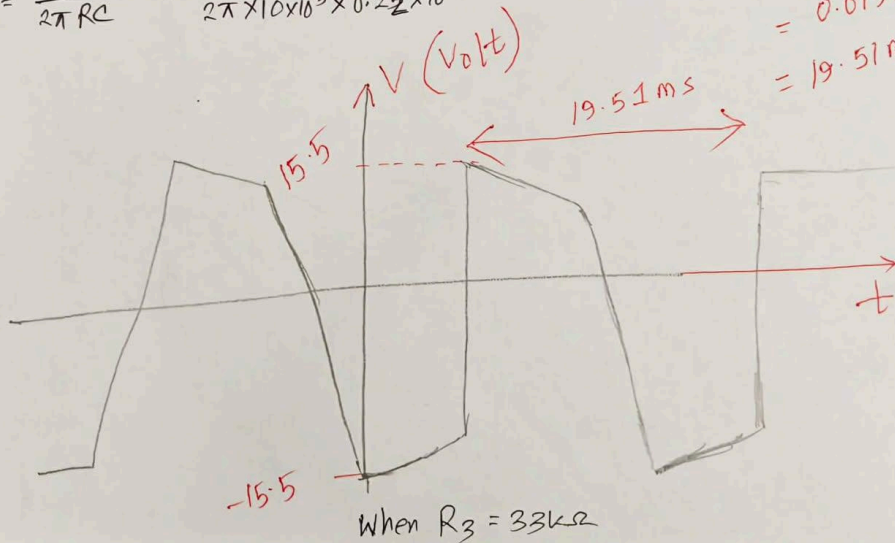
$$C_2 = 0.24\mu F$$

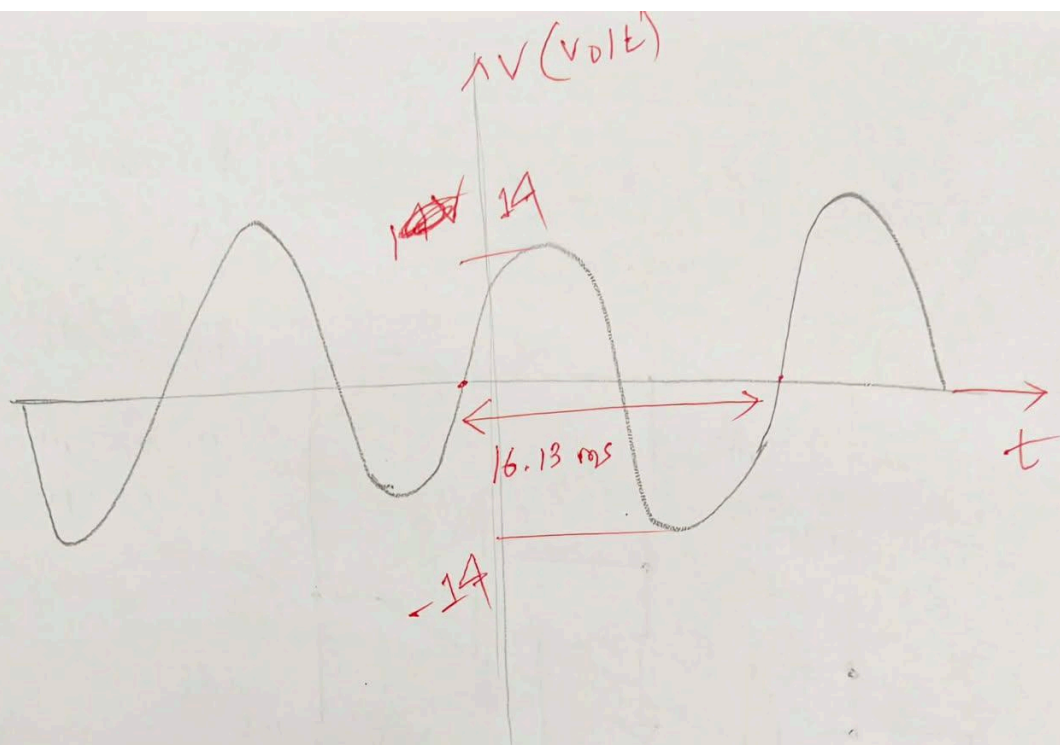


$$f = 51.24 \text{ Hz}$$

$$f = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 10 \times 10^3 \times 0.22 \times 10^{-6}} = 72.34 \text{ Hz}$$

$$T = \frac{1}{f} = \frac{1}{51.24} = 0.0195 \text{ s} = 19.51 \text{ ms}$$





$$R_B = 20 \text{ k}\Omega$$

$$f = 61.96 \text{ Hz}$$

$$T = 16.13 \text{ ms}$$

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16/04/2024

